quiz 卷 6Nov

Answer Key

IGCSE Physics 0625 · Thermal Physics

Section A — Multiple Choice

1 D 2 D 3 D 4 B 5 C

6 A 7 C 8 D 9 B 10 D

11 B 12 B 13 A 14 B 15 D

Section B — Structured Questions

Question 1 · Air in a car tyre · 8 marks

(a) [3] pV = constant → V = (1.02 × 1.0 × 10^5) ÷ (2.6 × 10^5) = 0.39 m^3

(b)(i) [3] separation: large / far apart; arrangement: irregular / random; motion: rapid / random (directions)

(b)(ii) [2] average kinetic energy of the particles decreases

Question 2 · Heating water in a saucepan · 9 marks

(a) [1] average kinetic energy of the particles increases / particles move faster

(b)(i) [2] ρ = m/V; 250 cm^3 = 2.5 × 10^{-4} m^3; m = ρV = 1000 × 2.5 × 10^{-4} = 0.25 kg

(b)(ii) [3] Δθ = 65 − 20 = 45 °C; E = mcΔθ = 0.25 × 4200 × 45 = 47 000 J

(b)(iii) [1] thermal energy is also transferred to the pan / surroundings

(c) [2] any two: takes longer to heat the water · more thermal energy needed for the same temperature rise · aluminium has a higher specific heat capacity